

ENT

This Knowledge Base (KB) acts as a **system-level instruction set** for the refinement pipeline. It prevents blind word-for-word translation and ensures context-aware, consistent, and academically appropriate output for content.

6	50+ Code Rules	40+ DNT Terms	5 Glossary Entries
---	-------------------	------------------	-----------------------

Context Mappings

TABLE OF CONTENTS

- 1. Transliteration vs. Translation — The Core Rule
- 2. Do-Not-Translate (DNT) Master List
- 3. Domain-Specific Glossary
- 4. Tone, Formality & Style Rules
- 5. Context Resolution Map
- 6. Code Block Preservation Rules
- 7. Quick Decision Flowchart
- 8. LLM Prompt Template

01 RULE 1 — TRANSLITERATION vs. TRANSLATION

■ The Golden Rule

Standard software engineering and computer science concepts must be **transliterated** (written in the local script retaining the English phonetic sound), **NOT** translated into obscure or unfamiliar native words that students will not recognise in real-world usage.

■ The Golden Rule

Standard software engineering and computer science concepts must be **transliterated** (written in the local script retaining the English phonetic sound), **NOT** translated into obscure or unfamiliar native words that students will not recognise in real-world usage.

When to Transliterate vs. When to Translate

Criterion	Transliterate ✓	Translate ✓
Is the English term used universally by working professionals?	Yes	No
Would a native translation obscure the meaning for students?	Yes	No
Is the term found verbatim in textbooks / docs / IDEs?	Yes	No
Is the term an acronym or brand-like (API, SQL, JSON)?	Yes	No
Is the term a common everyday word with a clear native equivalent?	No	Yes

Worked Examples

Object-Oriented Programming (Telugu)

Transliteration

■ ■■■■■-■■■■■ ■■■■■■■■■■■■ ✗ Literal translation — avoid

Database System (Hindi)

Transliteration

X Literal translation — avoid

Machine Learning (Tamil)

☒ Transliteration

■ ■ ■ ■ ■ ■ ■ ■ ■ ■ X Too obscure for practitioner use

■ *Note: This rule takes precedence over all general language translation guidelines. When in doubt, transliterate.*

02 RULE 2 — DO-NOT-TRANSLATE (DNT) MASTER LIST

The following items must appear **exactly as written** — in English, in the Latin alphabet — in the final translated output. They must never be transliterated or translated.

Programming Languages & Runtimes

Python	JavaScript	TypeScript	Java	C
C++	C#	Go	Rust	Ruby
PHP	Kotlin	Swift	R	MATLAB
Scala	Bash	Shell	PowerShell	Node.js
Deno	JVM	CLR		

Frameworks & Libraries

Flask	Django	FastAPI	React	Angular
Vue.js	Next.js	Spring Boot	TensorFlow	PyTorch
Keras	Scikit-learn	NumPy	Pandas	Matplotlib
Express.js	Laravel	Ruby on Rails	Bootstrap	Tailwind CSS

File Extensions & Config Files

.py	.js	.ts	.java	.cpp
.ipynb	.json	.yaml	.yml	.csv
.xml	.html	.css	.env	.sh
requirements.txt	Dockerfile	docker-compose.yml	package.json	.gitignore

Databases & Query Languages

SQL	MySQL	PostgreSQL	SQLite	MongoDB
Redis	Cassandra	Firebase	DynamoDB	Oracle DB
MS SQL Server	NoSQL	GraphQL	JDBC	ORM
Sequelize	Mongoose			

Cloud, DevOps & Infrastructure

AWS	GCP	Azure	Docker	Kubernetes
Terraform	Ansible	CI/CD	GitHub Actions	Jenkins
Helm	Nginx	Apache	Linux	Ubuntu
CentOS	Debian	macOS	Windows Server	

Protocols, Standards & Formats

HTTP	HTTPS	REST	gRPC	SOAP
WebSocket	TCP/IP	UDP	DNS	SSL
TLS	OAuth	JWT	API	JSON
XML	CSV	YAML	Markdown	UTF-8
ASCII	Base64			

System Commands & CLI Tools

pip	npm	yarn	git	curl
wget	ssh	scp	grep	awk
sed	chmod	sudo	apt	brew
conda	virtualenv			

■ Note: Any text inside backticks, code fences (``), or tags is automatically DNT. The LLM must preserve code blocks byte-for-byte.

03 RULE 3 — DOMAIN-SPECIFIC GLOSSARY

When a term is not on the DNT list, use the mapping below. **Consistency is mandatory** — the same English term must always produce the same output within a single course or document.

English Term	Action	Accepted Output / Note
Algorithm	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■
Array	Transliterate	■ ■ ■ ■ ■ / ■ ■ ■ / ■ ■ ■
Asynchronous	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Authentication	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ (Hindi OK)
Bandwidth	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Boolean	DNT	Boolean (never translate)
Buffer	Transliterate	■ ■ ■ ■ / ■ ■ ■
Cache / Caching	Transliterate	■ ■ ■ ■ / ■ ■ ■
Callback	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■
Class (OOP)	Transliterate	■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■
Cloud Computing	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Compiler	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■
Container	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■
Data Type	Translate OK	■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Debugging	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■
Deployment	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Encryption	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Exception / Error Handling	Transliterate	■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Foreign Key	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Framework	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Function / Method	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Hash / Hashing	Transliterate	■ ■ ■ ■ / ■ ■ ■
Inheritance (OOP)	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Instance	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Interpreter	Transliterate	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ / ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

English Term	Action	Accepted Output / Note
Iteration / Loop	Translate OK	■■■■■■■■■■ / ■■■■■■■■■■ (loop = ■■■■)
Latency	Transliterate	■■■■■■■■ / ■■■■■■■■
Library	Transliterate	■■■■■■■■ / ■■■■■■■■■■
Microservice	Transliterate	■■■■■■■■■■■■■■■■ / ■■■■■■■■■■■■■■■■
Migration (DB)	Transliterate	■■■■■■■■■■ / ■■■■■■■■■■
Module	Transliterate	■■■■■■■■■■ / ■■■■■■■■
Node (graph/network)	Transliterate	■■■■■ / ■■■■
Object (OOP)	Transliterate	■■■■■■■■■■■ / ■■■■■■■■■■
Overloading / Overriding	Transliterate	■■■■■■■■■■■■■■■■ / ■■■■■■■■■■■■
Package	Transliterate	■■■■■■■■■■■ / ■■■■■■
Parameter / Argument	Transliterate	■■■■■■■■■■■ / ■■■■■■■■■■
Pointer	Transliterate	■■■■■■■■■■■ / ■■■■■■■■
Primary Key	Transliterate	■■■■■■■■■ ■■ / ■■■■■■■■■■ ■■
Protocol	Transliterate	■■■■■■■■■■■■■■ / ■■■■■■■■■■■■
Query	Transliterate	■■■■■■■■■ / ■■■■■■■■
Queue	Transliterate	■■■■■ / ■■■■■
Recursion	Transliterate	■■■■■■■■■■■ / ■■■■■■■■
Repository	Transliterate	■■■■■■■■■■■■■■ / ■■■■■■■■■■■■
Runtime	Transliterate	■■■■■■■■■■■ / ■■■■■■■■
Schema	Transliterate	■■■■■■■■■ / ■■■■■■■■
Server	Transliterate	■■■■■■■■■ / ■■■■■■
Stack / Stack Trace	Transliterate	■■■■■■■■■ / ■■■■■■
String (data type)	Transliterate	■■■■■■■■■■■■■■ / ■■■■■■■■■■■■■ (NOT 'thread/rope')
Thread (OS)	Transliterate	■■■■■■■■■ / ■■■■■■■■ (NOT 'sewing thread')
Token (auth/NLP)	Transliterate	■■■■■■■■■ / ■■■■■■
Variable	Transliterate	■■■■■■■■■■■■■■ / ■■■■■■■■■■
Virtual Machine	Transliterate	■■■■■■■■■■■■■■■■■■■■■■■■ / ■■■■■■■■■■■■■■■■■■■■■■■■■
Wrapper	Transliterate	■■■■■■■■■ / ■■■■■■

■ *Note: When a term is followed by a language tag (e.g., 'Hindi OK'), a native-language equivalent is acceptable for that specific language but transliteration is still preferred for clarity.*

04 RULE 4 — TONE, FORMALITY & STYLE

Academic & Instructional Tone

- Use formal academic register throughout.
- Write as a university professor delivering a lecture, not a casual blogger.
- Avoid contractions, slang, colloquialisms, or informal particles.

Formal Pronouns (Language-Specific)

- Hindi: Use **मैं** (aap) — never **तुम** (tum) or **तू** (tu).
- Telugu: Use **నేను** (meeru) — never **నువ్వు** (nuvvu).
- Tamil: Use **நீ** (neengal) — never **நீ** (nee).
- Kannada: Use **ನಾನು** (neevu) — never **ನೀನು** (neenu).
- Bengali: Use **আমি** (apni) — never **তুমি** (tumi).

Clarity Over Linguistic Purity

- If a grammatically 'pure' translation of a technical term produces a word that students will not recognise in professional settings, **prefer the English transliteration**.
- Example: Prefer '**algorithm**' over an obscure Sanskrit-derived Telugu coinage.

Sentence Structure

- Preserve the original sentence meaning and information hierarchy.
- Do not split one concept across extra sentences or merge two explanations into one.
- Maintain bullet-point structure, numbered lists, and heading hierarchy exactly.

Brand & Product Names

- Treat software product names (VS Code, IntelliJ, Jupyter) as DNT.
- Translate surrounding prose; preserve the product name in the original Latin script.

Tone Comparison — Good vs. Bad

Scenario	Incorrect Style	Correct Style
Addressing the learner	हेलो फ्रेंड्स (informal)	प्रिय छात्र (formal)
Explaining a concept	basically यह एक box है (casual)	यह एक कॉन्टेनर है (academic)
Referring to a library	NumPy एक tool है	NumPy एक लैब्रेरी

05 RULE 5 — CONTEXT RESOLUTION MAP

Many English words carry multiple meanings. The LLM must always resolve ambiguous terms to their **Computer Science / IT meaning**. Use this table as the authoritative context map.

Term	Wrong Context (Avoid)	Correct CS Context
String	A piece of thread / rope	A sequence of characters — data type (e.g., 'hello')
Table	A piece of furniture	A structured data object in a relational database
Tree	A plant	A hierarchical data structure (binary tree, B-tree, etc.)
Queue	A line of people waiting	A FIFO data structure; also a message queue (RabbitMQ)
Stack	A pile of physical objects	A LIFO data structure; also a technology stack
Thread	A sewing thread / topic	A unit of execution in a process (OS threading)
Fork	A kitchen utensil	A system call that creates a child process; also git fork
Shell	An animal shell	A command-line interface (bash, zsh)
Kernel	A seed kernel	The core of an operating system
Monitor	A computer screen display	Observing system health / metrics (latency monitoring)
Port	A harbour for ships	A logical endpoint for network communication (port 8080)
Socket	An electrical socket	A network communication endpoint
Bridge	A physical bridge structure	A network device / Docker bridge network
Garbage	Physical waste / rubbish	Unused memory objects subject to garbage collection
Sandbox	A children's play sandbox	An isolated execution environment for testing
Cookie	A baked food item	A small data file stored in a browser by a web server
Session	A meeting / sitting	A server-side storage mechanism tracking user state
Pipeline	A physical pipe system	A sequence of data processing steps (CI/CD pipeline)
Registry	An official record book	A central configuration store (Docker Registry, Windows Registry)
Proxy	A legal representative	An intermediary server that forwards client requests

■ *Note: When the source text is ambiguous, assume the CS/IT context definition above. Never use the everyday/colloquial meaning when translating technical course content.*

06 RULE 6 — CODE BLOCK PRESERVATION

■ Absolute Rule — Code is Sacred

Any content identified as code **must be reproduced byte-for-byte**. Never alter, translate, transliterate, reformat, or paraphrase code. Treat code as a read-only artefact.

Code is identified by any of:

- Triple backtick fences: ```python ... ```
- Inline backticks: `variable\_name`
- HTML <code> or <pre> tags
- Indented blocks in Markdown (4-space indent)
- Jupyter cell inputs / outputs

Before / After Example

Source (English)	■ Correct Translation	■ Wrong Translation
<pre>**Step 3:** Run the following command: ```bash pip install flask python app.py --port 5000 ``` The server will start on port 5000.</pre>	<pre>**Step 3 (Telugu):** ■■■■■■ ■■■■■■ ■■■ ■■■■■■: ```bash pip install flask python app.py --port 5000 ``` ■■■■■■ ■■■■■■ 5000 ■■ ■■■■■■■■■■■■■■■■■■■■.</pre>	<pre>**Step 3 (WRONG):** ■■■■■■ ■■■■■■ ■■■ ■■■■■■: ```bash pip ■■■■■■■■■■ flask ← NEVER do this python ■■■■.py --■■■■■■■ 5000 ← NEVER do this ```</pre>

07 RULE 7 — QUICK DECISION FLOWCHART

Apply this decision tree to **every token / phrase** encountered during the refinement pass:



08 RULE 8 — LLM PROMPT TEMPLATE

Inject the following system prompt (along with this PDF as context) into your refinement API call. Replace {TARGET_LANGUAGE} and {COURSE_NAME} with dynamic values.

You are an expert technical translator specialising in Computer Science and Information Technology educational content. You are refining a machine-translated draft from English into {TARGET_LANGUAGE} for the course: {COURSE_NAME}. MANDATORY RULES – follow in strict priority order: 1. CODE PRESERVATION: Any text inside code blocks (`` ` ``), inline backticks (`), or tags must be reproduced CHARACTER-FOR-CHARACTER. Never alter code. 2. DNT LIST: All terms in the Do-Not-Translate list must remain in the original English Latin alphabet. Refer to Section 2 of the attached Knowledge Base PDF. 3. TRANSLITERATION OVER TRANSLATION: Prefer phonetic transliteration in the target script over literal translation for all standard CS/IT terminology. Refer to Section 1 and Section 3 (Glossary) of the Knowledge Base PDF. 4. CONTEXT: Assume ALL content is within the domain of Computer Science and IT. Resolve ambiguous words using Section 5 (Context Resolution Map). 5. TONE: Use formal, academic register. Apply formal second-person pronouns as specified in Section 4. Never use colloquialisms or slang. 6. CONSISTENCY: Use the exact same output for identical input terms throughout the entire document. Input (Machine Translation Draft): --- {DRAFT_TRANSLATION} --- Produce the refined translation in {TARGET_LANGUAGE} only. Do not add explanations.

API Integration Checklist

- Attach this Knowledge Base PDF via the File API (Gemini) or inject text into the system prompt.
- Set temperature = 0.1 for maximum consistency across batch translations.
- Set max_output_tokens to at least 2× the input length to prevent truncation.
- Process one section / chapter at a time to stay within context window limits.
- Run a consistency post-pass: verify that all DNT terms appear unchanged.
- Log any terms the LLM changed unexpectedly and update this KB accordingly.